

Power Flow Security Maximization via Inverse Chance Constrained Optimization

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Abstract—The chance-constrained optimal power flow (CC-OPF) essentially finds the low-cost generation dispatch scheme ensuring operational constraints are met with a specified probability, termed the security level. While the security level is a crucial input parameter, how it shapes the CC-OPF feasibility boundary has not been revealed. Changing the security level from a parameter to a decision variable, this letter proposes a security maximization approach based on the chance constrained DC-OPF model, termed inverse CC-OPF (ICC-OPF), that seeks the maximum security level achievable by the system. To efficiently solve the ICC-OPF, we design a Newton-Raphson-like iteration algorithm leveraging the duality-based sensitivity analysis of an associated surrogate problem. Numerical experiments validate the proposed approach, revealing complex feasibility boundaries for security levels that underscore the importance of coordinating security levels across multiple chance constraints.

Index Terms—Chance constraint, optimal power flow, duality, sensitivity, Newton-Raphson iteration.

I. INTRODUCTION

CHANCE-CONSTRAINED optimal power flow (CC-OPF) is a common framework for managing uncertainties in power systems [1]. It incorporates uncertain factors $\mathbf{w}$ (e.g., wind or solar power) modeled as random variables and seeks decision variables $\mathbf{x}$ (e.g., generation dispatch) that satisfy operational security constraints with high probabilities, i.e., $\Pr\{g(\mathbf{x}, \mathbf{w}) \leq 0\} \geq \beta$, where β is the *security level* that serves as a key input parameter conventionally pre-specified by the system operator. The value of β is often chosen based on heuristics or

limited sensitivity analyses (e.g., ranging from 80% to 99.99% in [2], [3], [4]), which makes the resulting CC-OPF solutions are inherently tied to the adopted security level.

If we change the role of the security level from an input parameter to a decision variable, a fundamental question naturally arises: what is the maximum security level a power system can achieve under given uncertainties? This leads to a new problem formulation focused on security maximization.

To the authors' knowledge, this question has not been systematically studied in the literature. The existing CC-OPF research has primarily focused on deriving tractable deterministic reformulations for chance constraints, for instance, the Polynomial Chaos Expansion (PCE) technique is adequate for parametric uncertainties with known distributions [5], [6], the Gaussian mixture model (GMM) can handle multi-modal distributions and transforms the original problem into mixed-integer conic programs [7], [8], and distributionally robust optimization (DRO) describes the uncertainty in probability distribution by ambiguity sets and leads to convex reformulations [4]. In case of linear chance constraint (i.e., $g(\mathbf{x}, \mathbf{w}) = \mathbf{w}^T \mathbf{x} - b$) which is the most common and simple model, a canonical result of these methods is the second-order cone (SOC) constraint $\tilde{\mathbf{w}}^T \mathbf{x} + \psi_\beta \|\Sigma^{\frac{1}{2}} \mathbf{x}\|_2 \leq b$. This form incorporates a safety margin by accounting for the uncertain parameters' covariance matrix Σ and a safety factor ψ_β , which is derived from a pre-defined security level β serving as a required input for all the aforementioned methods.

A parallel line of study focuses on adaptively tuning or updating the safety factor ψ_β to satisfy a target security level β , particularly when the uncertainty distribution is unknown or inaccurate. This includes methods based on empirical evaluation [9], data-driven approaches [10], distributionally robust optimization [11], [12], and stochastic model predictive control methods evolving closed-loop adjustment [13], [14] and distribution-free adaptation [15]. Although these methods tune the safety factor ψ_β , their fundamental goal is to ensure compliance with a pre-defined target β . The security level β is still treated as an exogenous input rather than a variable.

Another related stream of research is inverse optimization, which focuses on inferring the unknown parameters of an optimization problem from historical optimal solutions [16]. For instance, [17] studies an inverse problem for distributionally robust CC-OPF, effectively revealing an operator's subjective security preferences from historical economic dispatch data.

Overall, the existing research has revealed the importance of security level and a naive approach to setting security level is likely to cause infeasibility of CC-OPF. But there

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is a lack of systematic approaches to evaluate the maximum security level that a power system can achieve under uncertainty.

Our work addresses this gap by proposing a problem that *inverts* the conventional role of β in the DC-based CC-OPF formulation, treating it as a decision variable to be maximized. Specifically, this letter establishes the inverse CC-OPF (ICC-OPF) as a security-oriented variant added to the conventional CC-OPF framework, that finds the maximum security level along certain security increase directions and figures out those critical lines as the system bottleneck that need to be prioritized. To enable an efficient solution method, we propose a Newton-Raphson-like (NR-like) iteration algorithm by leveraging the duality-based sensitivity of the optimal value of a surrogate problem associated with the ICC-OPF. For the primary case of Gaussian uncertainty, we derive a convex surrogate problem and a closed-form NR-like iteration scheme which is computationally friendly, and we outline its extension to non-Gaussian cases. From the numerical experiments on two IEEE test systems, the ICC-OPF provides new insights into how the maximum security levels of different chance constraints are restricting each other, which serves as a useful complement to the conventional CC-OPF.

The rest of this letter is organized as follows. Section II introduces our ICC-OPF formulation. Section III develops a solution algorithm based on a surrogate model, sensitivity analysis and NR-like iteration framework. Section IV presents numerical experiments on IEEE test systems to validate the proposed method, with visualizations and practical applications. Section V concludes this letter.

II. PROBLEM FORMULATION

A. A Quick Review of CC-OPF

The single-period CC-OPF problem under the DC power flow model is formulated as

$$\min_{\bar{\mathbf{P}}_g, \alpha_g} \mathbb{E}\{\mathbf{P}_g^T \mathbf{c}_2 \mathbf{P}_g + \mathbf{c}_1^T \mathbf{P}_g\} \quad (1a)$$

$$s.t. \quad \mathbf{P}_g = \bar{\mathbf{P}}_g - \alpha_g \mathbf{1}_n^T \left[(\mathbf{P}_w - \bar{\mathbf{P}}_w) - (\mathbf{P}_d - \bar{\mathbf{P}}_d) \right] \quad (1b)$$

$$\mathbf{1}_n^T (\mathbf{P}_g + \mathbf{P}_w - \mathbf{P}_d) = 0 \quad (1c)$$

$$\mathbf{f} = \mathbf{T}_f (\mathbf{P}_g + \mathbf{P}_w - \mathbf{P}_d) \quad (1d)$$

$$\mathbf{1}_n^T \alpha_g = 1 \quad (1e)$$

$$\alpha_{g,i} = 0, \quad \bar{P}_{g,i} = 0, \quad \forall i \notin \mathcal{V}_g \quad (1f)$$

$$\alpha_g \geq \mathbf{0} \quad (1g)$$

$$\Pr\{P_{g,i} \leq P_{g,i}^{\max}\} \geq \beta_i, \quad \forall i \in \mathcal{V}_g \quad (1h)$$

$$\Pr\{P_{g,i} \geq P_{g,i}^{\min}\} \geq \beta_i, \quad \forall i \in \mathcal{V}_g \quad (1i)$$

$$\Pr\{f_{ij} \leq f_{ij}^{\max}\} \geq \beta_{ij}, \quad \forall (i, j) \in \mathcal{E} \quad (1j)$$

$$\Pr\{f_{ij} \geq -f_{ij}^{\max}\} \geq \beta_{ij}, \quad \forall (i, j) \in \mathcal{E} \quad (1k)$$

which for a power system with buses $\mathcal{V}$ and lines $\mathcal{E}$ determines two decision variables: the base-case dispatch $\bar{\mathbf{P}}_g \in \mathbb{R}^{|\mathcal{V}|}$ and participation factors $\alpha_g \in \mathbb{R}^{|\mathcal{V}|}$ for generators in subset $\mathcal{V}_g \subseteq \mathcal{V}$. These decision variables form an affine control policy (1b) that governs the actual power output $\mathbf{P}_g$ in real-time, adapting the base-case dispatch $\bar{\mathbf{P}}_g$ to the deviations in uncertain renewable generation $\mathbf{P}_w$ and loads $\mathbf{P}_d$ from their forecasts $(\bar{\mathbf{P}}_w, \bar{\mathbf{P}}_d)$, with α_g serving as the control coefficients.

As the actual output $\mathbf{P}_g$ is consequently stochastic, the objective function (1a) is formulated to minimize the expected quadratic generation cost.

The vector $\mathbf{f} \in \mathbb{R}^{|\mathcal{E}|}$ denotes the power flow on transmission lines. Constraints (1c)-(1d) represent the DC power flow model, where (1c) enforces the system-wide power balance, and (1d) maps the power injections to line flows via the Power Transfer Distribution Factor (PTDF) matrix $\mathbf{T}_f$. Constraints (1e)-(1g) impose operational limits on the decision variables α_g and $\mathbf{P}_g$.

The system security under uncertainty is enforced by chance constraints (1h)-(1k), with the probability operators $\Pr\{\cdot\}$. In particular, (1h)-(1i) ensure that each generator output $P_{g,i}$ satisfies the upper and lower bounds $P_{g,i}^{\max}$ and $P_{g,i}^{\min}$ with probability levels no less than the *pre-defined security levels* β_i . Similarly, (1j)-(1k) ensure that line flows are within their respective limits with probability levels no less than β_{ij} .

For simplicity, we henceforth write (1) into the following compact form:

$$\min_{\mathbf{x}} C(\mathbf{x}) \quad (2a)$$

$$s.t. \quad \mathbf{A}_1 \mathbf{x} = \mathbf{b}_1, \quad \mathbf{A}_2 \mathbf{x} \leq \mathbf{b}_2 \quad (2b)$$

$$\Pr\{\mathbf{w}_k^T \mathbf{x} \leq d_k\} \geq \beta_k, \quad \forall k \in \mathcal{V}_g \cup \mathcal{E} \quad (2c)$$

where $\mathbf{x}$ denotes the concatenated vector of decision variables $\bar{\mathbf{P}}_g$ and α_g (the line flow notation $\mathbf{f}$ is eliminated as it can be linearly expressed by $\bar{\mathbf{P}}_g$ and α_g); (2b) represents deterministic equality constraints (1c)-(1f) and deterministic inequality constraints (1g); and (2c) represents the probabilistic chance constraints (1h)-(1k), with some entries of vector $\mathbf{w}_k$ following certain probability distributions (which comes from the uncertainty in $\mathbf{P}_w$ and $\mathbf{P}_d$) and β_k being a pre-defined security level for respective line flows and generation outputs; $\mathbf{A}_1, \mathbf{A}_2, \mathbf{b}_1$ and $\mathbf{b}_2$ are the associated coefficient matrices.

B. Formulating the ICC-OPF

In (2) the security level β_k is usually predefined by engineering heuristics. However, when the uncertain parameter $\mathbf{w}_k$ follow long-tail distributions and β_k is set sufficiently close to 1, the chance constraints need to accommodate extreme events at the tails of the uncertainty distribution, which will make the CC-OPF infeasible as it exceeds the physical limits or operational capabilities of the power system. Therefore, it is of interest to reconsider the CC-OPF problem in an inverse manner. Instead of a predefined constant β_k , we set β_k as a variable and find the maximum security level under which problem (2) maintains feasible, aiming to characterize the highest security level that the system can achieve.

The points of maximum security levels locate at the feasibility boundary surface of the CC-OPF which can be reached from a starting point along a certain direction. Thus, we propose the formation of the ICC-OPF problem as follows:

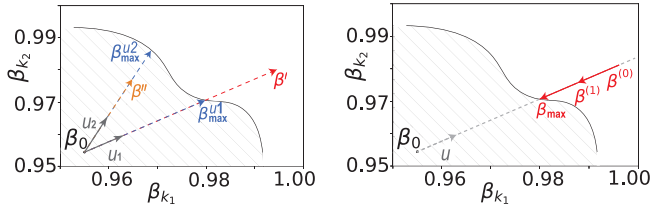
$$\beta_{\max} = \max_{\mathbf{x}, \beta} \beta \quad (3a)$$

$$s.t. \quad \mathbf{A}_1 \mathbf{x} = \mathbf{b}_1, \quad \mathbf{A}_2 \mathbf{x} \leq \mathbf{b}_2 \quad (3b)$$

$$\Pr\{\mathbf{w}_k^T \mathbf{x} \leq d_k\} \geq \beta_k, \quad \forall k \in \mathcal{V}_g \cup \mathcal{E} \quad (3c)$$

$$\beta_k = \beta u_k + \beta_{0,k}, \quad \forall k \in \mathcal{V}_g \cup \mathcal{E} \quad (3d)$$

where (3d) sets the search direction in the security level space starting from a given offset point $\beta_0 = [\beta_{0,k}]$ along



(a) Geometric Interpretation of (3d) (b) Geometric Interpretation of Alg 1

Fig. 1. Geometric Interpretation of the ICC-OPF problem.

a fixed unit direction vector $\mathbf{u} = [u_k]$.¹ Fig. 1(a) provides a geometric interpretation in the β_{k_1} - β_{k_2} plane. Starting from the offset point $\beta_0 = (\beta_{0,k_1}, \beta_{0,k_2})$, a security level vector $\beta = (\beta_{k_1}, \beta_{k_2})$ can be represented using the specified direction $\mathbf{u}$ (the gray arrow) and a scalar decision variable β measuring its modulus. The feasible region is shown with gray hatching. A significantly large β' (red arrow) leads to infeasibility, whereas a small β'' (yellow arrow) remains within the feasible region. The optimal value $\beta_{\max}$ lies precisely on the boundary of the feasible region. For any given direction $\mathbf{u}$, a unique optimal value $\beta_{\max}$ exists, as indicated by the blue arrow.

The solution of (3), denoted by $\beta_{\max}$, represents the largest feasible security increment along the direction $\mathbf{u}$. Collecting the optima of ICC-OPF under a number of directions $\mathbf{u}$ gives the feasibility boundary capturing the trade-offs and interdependencies among the security levels of different line flows, which will be demonstrated in the case studies.

III. SOLUTION METHOD

A. The Surrogate Model With Slack Variables

The primary challenge in solving the ICC-OPF problem (3) is the non-convexity induced by treating the security level β as a decision variable. This structure creates a complex coupling between the variables $\mathbf{x}$ and β in the chance constraints (3c).

To address this challenge, we propose a surrogate model that bypasses the complexity in (3). The surrogate model is a parametric optimization with respect to β . For a given value of β , slack variables $\mathbf{s}$ are introduced to quantify the possible violation of the chance constraints:

$$\mathbf{s}^*(\beta) = \arg \min_{\mathbf{x}, \mathbf{s}} \|\mathbf{s}\|_2 \quad (4a)$$

$$\text{s.t.} \quad (3b), (3d) \quad (4b)$$

$$\Pr\{\mathbf{w}_k^T \mathbf{x} - s_k \leq d_k\} \geq \beta_k : \lambda_k, \forall k \quad (4c)$$

$$s_k \geq 0, \forall k \in \mathcal{V}_g \cup \mathcal{E} \quad (4d)$$

where $\lambda_k \geq 0$ represents the dual variable associated with constraint (4c). For the surrogate problem, the optimal solution $\mathbf{s}^*$ is an implicit function of β , and we have the following result that links the surrogate model to ICC-OPF.

Theorem 1: A security level $\hat{\beta}$ is the optimum $\beta_{\max}$ of the ICC-OPF problem (3), if and only if the optimum of problem (4) satisfies $\|\mathbf{s}^*(\hat{\beta})\|_2 = 0$ and $\|\mathbf{s}^*(\hat{\beta} + \varepsilon_\beta)\|_2 > 0$ for any arbitrarily small $\varepsilon_\beta > 0$.

¹The direction vector $\mathbf{u}$ is constructed to be sparse in practice, allowing operators to prioritize security characterizations and enhancements for critical constraints. Such constraints can be identified a priori, for instance, by inspecting the dual variables with respect to the security constraints of a conventional CC-OPF.

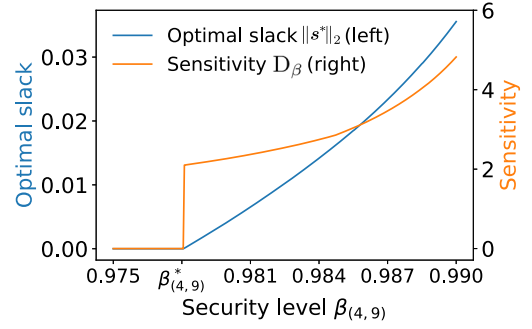


Fig. 2. Illustration of the relationship between $\|\mathbf{s}^*\|_2$ and D_β with respect to β . For better interpretability, the horizontal axis is linearly transformed as $\beta_{(4,9)} = \sqrt{2}\beta + 0.95$. The figure is plotted for the IEEE 14-bus system with parameters identical to those in Section IV.

Proof (Sufficiency): Assume $\|\mathbf{s}^*(\hat{\beta})\|_2 = 0$ and $\|\mathbf{s}^*(\hat{\beta} + \varepsilon_\beta)\|_2 > 0$ for any $\varepsilon_\beta > 0$. The first condition, $\|\mathbf{s}^*(\hat{\beta})\|_2 = 0$, implies all slack variables $s_k^* = 0$, which means the original chance constraints (3c) are satisfied at $\beta = \hat{\beta}$. Hence, $\hat{\beta}$ is a feasible solution to the ICC-OPF problem. The second condition, $\|\mathbf{s}^*(\hat{\beta} + \varepsilon_\beta)\|_2 > 0$ for any $\varepsilon_\beta > 0$, implies that for any $\beta' = \hat{\beta} + \varepsilon_\beta$ greater than $\hat{\beta}$, at least one of the constraints (3c) must be violated (otherwise, $\|\mathbf{s}^*(\hat{\beta} + \varepsilon_\beta)\|_2 = 0$ would hold, a contradiction), rendering that β' is infeasible. Therefore, $\hat{\beta} = \beta_{\max}$.

(Necessity): Assume $\hat{\beta} = \beta_{\max}$. By definition, $\hat{\beta}$ is a feasible solution to the ICC-OPF problem, implying $\mathbf{s} = \mathbf{0}$ is a feasible solution to problem (4). Thus $\|\mathbf{s}^*(\hat{\beta})\|_2 = 0$. Now, assume for the sake of contradiction that there exists an $\varepsilon_\beta > 0$ such that $\mathbf{s}^*(\hat{\beta} + \varepsilon_\beta) = \mathbf{0}$. This would imply that $\hat{\beta} + \varepsilon_\beta$ is a feasible solution to the ICC-OPF, which contradicts the premise that $\hat{\beta} = \beta_{\max}$ is the maximum. Thus, $\|\mathbf{s}^*(\hat{\beta} + \varepsilon_\beta)\|_2 > 0$ must hold for any $\varepsilon_\beta > 0$. ■

This theorem implies that the search for $\beta_{\max}$ is equivalent to finding the point where the optimal value of the corresponding surrogate problem $\|\mathbf{s}^*\|_2$ changes from 0 to 0^+ , as illustrated by the blue curve in Fig. 2. Utilizing the property of $\mathbf{s}^*(\beta)$ of the surrogate model (4) at any fixed value of β , an NR-like iteration scheme will be designed to efficiently move β to the maximal feasible security level $\beta_{\max}$, as presented in the following subsection.

B. NR-Like Iteration Scheme

Due to the nonlinear relationship between $\mathbf{s}^*$ and β , an NR-like iteration scheme is designed to find $\beta_{\max}$ such that $\|\mathbf{s}^*(\beta_{\max})\|_2 = 0$. The iteration process starts with a sufficiently large β (such that $\beta > \beta_{\max}$ and consequently, $\|\mathbf{s}^*\|_2 > 0$) and gradually decreases β until the corresponding $\|\mathbf{s}^*\|_2$ approaches zero. This procedure identifies the aforementioned critical point at which $\|\mathbf{s}^*\|_2 = 0$ but any slight increase in β would result in $\|\mathbf{s}^*\|_2 > 0$. To construct the NR-like iteration scheme, let us begin with the first-order expansion of $\|\mathbf{s}^*\|_2$ with respect to β (suppose $\beta > \beta_{\max}$):

$$\|\mathbf{s}^*\|_2 = \|\mathbf{s}^*\|_2|_{\beta_0} + D_\beta|_{\beta_0}(\beta - \beta_0) + \mathcal{O}((\beta - \beta_0)^2), \quad (5)$$

where

$$D_\beta = d\|\mathbf{s}^*\|_2/d\beta \quad (6)$$

denotes the sensitivity of the optimal objective value of (4) with respect to parameter β . Thus, the iteration formula for computing β towards $\|s^*\|_2 = 0$ is:

$$\beta^{(t+1)} = \beta^{(t)} - \eta_t \|s^*\|_2^{(t)} / D_\beta^{(t)}, \quad (7)$$

where $\beta^{(t)}$ denotes the value of β at t -th iteration, and η_t is the step size. By [18, Th. 8.2], the sensitivity D_β is given by the partial derivative of the Lagrangian $\mathcal{L}$ with respect to β at the primal-dual optimal solution $(\mathbf{x}^*, s^*, \lambda^*)$ to the surrogate problem (4):

$$\begin{aligned} D_\beta &= \frac{\partial}{\partial \beta} \mathcal{L}(\mathbf{x}^*, s^*, \lambda^*, \beta) \\ &= \frac{\partial}{\partial \beta} \left(\sum_{k \in \mathcal{V}_g \cup \mathcal{E}} \lambda_k^* (\beta_k - \Pr\{\mathbf{w}_k^T \mathbf{x}^* - s_k^* \leq d_k\}) \right), \end{aligned} \quad (8)$$

which captures both explicit dependence via β_k and implicit effects through optimal variables $\lambda^*, \mathbf{x}^*$ and s^* .

We have the following result for checking the optimality of ICC-OPF in terms of $s^*(\beta)$ and D_β :

Proposition 1: At the primal-dual optimal solution to surrogate problem (4), we have $\beta = \beta_{\max}$ if and only if all of the following conditions are met:

- 1) $\|s^*(\beta)\|_2 = 0$;
- 2) $D_\beta^+ > 0$, $D_\beta^- = 0$;

where D_β^+ and D_β^- denote the right-hand and left-hand derivatives of $\|s^*(\beta)\|_2$ with respect to β , respectively.

Proof: Condition 1 follows directly from Theorem 1, as $\beta_{\max}$ is a feasible solution to the ICC-OPF problem.

To prove Condition 2, consider the right-hand derivative for $\|s^*\|_2$ at $\beta_{\max}$:

$$D_\beta^+ = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} (\|s^*(\beta_{\max} + \varepsilon)\|_2 - \|s^*(\beta_{\max})\|_2). \quad (10)$$

From Theorem 1, we know $\|s^*(\beta_{\max})\|_2 = 0$ and $\|s^*(\beta_{\max} + \varepsilon)\|_2 > 0$ for any $\varepsilon > 0$. The expression simplifies to $\lim_{\varepsilon \rightarrow 0^+} (\|s^*(\beta_{\max} + \varepsilon)\|_2 / \varepsilon)$, which must be greater than zero. Also it is trivial to claim that D_β is identically zero for any $\beta < \beta_{\max}$, and its left-hand derivative at $\beta_{\max}$, D_β^- , is consequently zero. ■

As the NR-like algorithm searches β decreasing from a large initial value, $\beta_{\max}$ is exactly the point where the sensitivity D_β jumps from a positive value to zero (see the orange curve in Fig. 1). Therefore, $\beta_{\max}$ can be numerically identified by checking the conditions:

$$\|s^*(\beta)\|_2 \leq \varepsilon_s \text{ and } D_\beta > \varepsilon_D, \quad (11)$$

where ε_s and ε_D are small positive values serving as the prescribed tolerances for the slack variable and sensitivity, respectively. Considering that the probability term in (9) is still implicit, the next subsection will derive an analytical expression of D_β under the Gaussian uncertainty.

C. Analytical Formulation Under Gaussian Uncertainty

Assume that the uncertain parameter $\mathbf{w}_k$ follows a Gaussian distribution, i.e., $\mathbf{w}_k \sim \mathcal{N}(\bar{\mathbf{w}}_k, \Sigma_k)$, where $\bar{\mathbf{w}}_k$ and Σ_k denote the mean and covariance matrix, respectively. This assumption is commonly adopted in the existing literature [1], [2] as a simplification of the complex real situations. A discussion on the relaxation to non-Gaussian uncertainty will be given

in the next subsection. Under the Gaussian uncertainty, constraint (4c) can be explicitly reformulated as [1]:

$$\bar{\mathbf{w}}_k^T \mathbf{x} - s_k + \Phi^{-1}(\beta_k) \|\Sigma_k^{\frac{1}{2}} \mathbf{x}\|_2 \leq d_k : \lambda_k, \quad \forall k \in \mathcal{V}_g \cup \mathcal{E} \quad (12)$$

where Φ^{-1} denotes the inverse cumulative distribution function (ICDF) of the standard Gaussian distribution. Problem (4) becomes a second-order cone program (SOCP) when (4c) is transformed to (12), which can be efficiently solved by the off-the-shelf solvers and the corresponding optimal value of the dual variable λ_k^* obtained simultaneously as a byproduct.

Substituting the chance constraint expression in (12) into (9), and applying the chain rule, we obtain

$$D_\beta = \sum_{k \in \mathcal{V}_g \cup \mathcal{E}} \frac{\partial \|s^*\|_2}{\partial \phi_k} \frac{d\phi_k}{d\beta}, \quad (13)$$

where $\phi_k = \Phi^{-1}(\beta_k)$, and

$$\frac{\partial \|s^*\|_2}{\partial \phi_k} = \lambda_k^* \|\Sigma_k^{\frac{1}{2}} \mathbf{x}^*\|_2, \quad \forall k \in \mathcal{V}_g \cup \mathcal{E}. \quad (14)$$

According to the definition of the ICDF Φ^{-1} , we have

$$\frac{d\phi_k}{d\beta} = \frac{d\phi_k}{d\beta_k} u_k = \sqrt{2\pi} u_k \exp\left(-\frac{\phi_k^2}{2}\right). \quad (15)$$

Substituting (14), (15) into (13) yields the computationally friendly formula for D_β :

$$D_\beta = \sqrt{2\pi} \sum_{k \in \mathcal{V}_g \cup \mathcal{E}} u_k \lambda_k^* \|\Sigma_k^{\frac{1}{2}} \mathbf{x}^*\|_2 \exp\left(-\frac{\phi_k^2}{2}\right). \quad (16)$$

Using (16), the NR-like iteration scheme in (7) can be executed with a step size η_t . Note that all $\lambda_k^* = 0$ when $\beta < \beta_{\max}$, which results in the zero value of D_β , i.e., a loss of gradient. To address this issue, a dynamic step size η_t is designed: starting from $\eta_t = 1$, if the updated β leads to an interior point of the ICC-OPF feasible region ($\beta < \beta_{\max}$), we perform a rollback and halve the step size η_t , so that each accepted β is an exterior point ($\beta > \beta_{\max}$). The iteration terminates when the β is sufficiently close to $\beta_{\max}$ from the right-hand side of $\beta_{\max}$. A geometric interpretation is provided in Fig. 1(b), and the detailed algorithm is presented in Algorithm 1.

D. Extension to Non-Gaussian Uncertainty

The proposed algorithm can be extended to non-Gaussian uncertainties by combining the following techniques.

First, in case that the non-Gaussian uncertainties admit a Polynomial Chaos Expansion (PCE) [5], the uncertain term $\mathbf{w}^T \mathbf{x}$ is approximated by a truncated PCE, $\hat{g}_L(\mathbf{x}, \xi) = \sum_{i=0}^L c_i(\mathbf{x}) \Psi_i(\xi)$, where $\{\Psi_i\}$ are orthogonal polynomial bases. The chance constraints are then conservatively approximated by deterministic SOC constraints using the mean and variance of $\hat{g}_L$ [6]:

$$c_{0,k}(\mathbf{x}) - s_k + \phi_k \left[\sum_{i=1}^L c_{i,k}(\mathbf{x})^2 \right]^{1/2} \leq d_k, \quad \forall k. \quad (17)$$

Since this formulation is structurally identical to the Gaussian case (12), the sensitivity D_β required by our framework can be computed analogously, and the convergence properties are maintained in this approximated system.

Second, we can also express a non-Gaussian PDF by the GMM in the form of $\sum_j \pi_j \mathcal{N}(\mu_j, \Sigma_j)$, where $\mathcal{N}(\mu_j, \Sigma_j)$

Algorithm 1 NR-Like Solution Method for ICC-OPF

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1: Set tolerances  $\varepsilon_s, \varepsilon_D$ , iteration index  $t \leftarrow 0$ , maximum
   number of iterations  $T$ , and flag  $flag_\eta \leftarrow \text{True}$ 
2: Initialize a sufficiently large  $\beta^{(t)}$  such that  $\|s^{*(t)}(\beta)\|_2 > 0$ 
3: Solve the surrogate problem (4) in its SOCP form with
    $\beta^{(t)}$  to obtain  $s^{*(t)}, \lambda^{*(t)}$  and  $D_\beta^{(t)}$ 
4: while  $t \leftarrow t + 1 < T$  and  $\|s^*\|_2 > \varepsilon_s$  do
5:   if  $flag_\eta$  then
6:      $\eta_t \leftarrow 1$ 
7:   end if
8:   Update the estimated value  $\hat{\beta} \leftarrow \beta - \eta_t \|s^*\|_2 / D_\beta$ 
9:   Solve (4) with  $\hat{\beta}$  to obtain  $\hat{s}^*, \hat{\lambda}^*$ , and  $\hat{D}_\beta$ 
10:  if  $|\hat{D}_\beta| \leq \varepsilon_D$  then
11:    Set  $\eta_t \leftarrow \eta_t / 2, flag_\eta \leftarrow \text{False}$ 
12:  else
13:    Update  $\beta \leftarrow \hat{\beta}, D_\beta \leftarrow \hat{D}_\beta, s^* \leftarrow \hat{s}^*$ 
14:    Set  $flag_\eta \leftarrow \text{True}$ 
15:  end if
16: end while
17: return  $\beta$ 

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refers to the j -th Gaussian component. Using the risk allocation technique [7], the overall chance constraint $\Pr\{\cdot\} \geq \beta$ is decomposed into a set of individual Gaussian chance constraints, $\Pr\{\cdot\} \geq p_j$, for each Gaussian component. The ICC-OPF is then reformulated as maximizing the total security level $\beta = \sum_j \pi_j p_j$, which can be similarly solved using our NR-like scheme, relying on the sensitivities of the surrogate SOCP problem.

Third, for generic, closed-box uncertainty models where an analytical SOCP reformulation is unavailable, the sensitivity D_β can be estimated by applying numerical perturbation (i.e., finite differences) to the surrogate model (4) so that our algorithm can still be executed.

IV. CASE STUDY

The proposed method is verified on the IEEE 14- and 39-bus systems. Uncertainty is introduced via renewable generators at select buses, modeled as Gaussian injections with a randomly generated positive semi-definite covariance matrix Σ , scaled such that its diagonal elements are 0.1.

For the IEEE 14-bus system, renewable generators are placed at buses 1, 3, 6, and 9. To push the system closer to its feasibility boundary, the loads are set to twice their original values from MATPOWER [19], while the maximum flow limit $f_{ij}^{\max}$ is set to 12.5% of the line's susceptance b_{ij} . For the IEEE 39-bus system, renewable generators are placed at buses 3, 8, 16, and 26, following the same distribution. Loads are scaled by 1.25, and $f_{ij}^{\max}$ is set to $0.1b_{ij}$. Additionally, for those single lines connecting generators to the network, we set $f_{ij}^{\max} = 0.2b_{ij}$.

The tolerances for the algorithm are $\varepsilon_s = \varepsilon_D = 10^{-6}$. A valid starting point requires an initial security level $\beta^{(0)}$, which is an exterior point of the ICC-OPF (i.e., $\|s^*(\beta^{(0)})\|_2 > 0$), which we construct by setting one security level $\beta_{k_1}^{(0)}$ to a high value of 0.999 and the initial value $\beta^{(0)} = (0.999 - \beta_{0,k_1})/u_{k_1}$. This initialization method proved effective in all subsequent experiments.

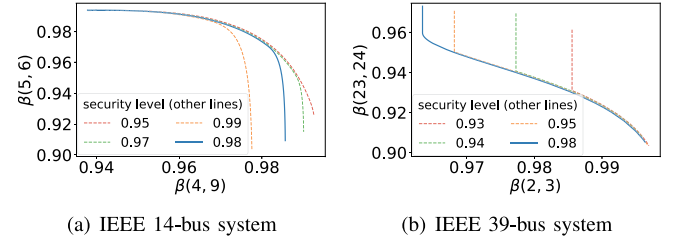


Fig. 3. Feasibility boundaries in the space of security levels.

A. Verification of NR-Like Method

The performance of Algorithm 1 can be characterized by the accuracy of the sensitivity D_β and optimum $\beta_{\max}$, which is verified by the following perturbation tests.

In the IEEE 39-bus system, set $\beta_i = 0.95, i \in \mathcal{V}_g$, generate $\beta_{ij}, (i, j) \in \mathcal{E}$ using (3d) with $\beta_{0,k} = 0.95$, and set $u_{ij} = \sqrt{2}/2$ for lines (2, 3) and (23, 24) and zero for the remaining lines. The analytical sensitivity computed by (16) is 12.5493, which closely matches the numerical perturbation result 12.2301 (step size 10^{-3}), with a relative error of approximately 2.5%, validating the accuracy of formula (16).

Our NR-like algorithm finds the solution with superlinear convergence in about 5 iterations, significantly outperforming a naive trial-and-error search that requires about 19 iterations. To validate optimality of the obtained $\beta_{\max}$, evaluate the feasibility of the original CC-OPF (2) at $\beta_{\max}$ and at $\beta_{\max} + 10^{-5}$. It turns out that the former is feasible and the latter is not, verifying that $\beta_{\max}$ lies sufficiently close to the feasibility boundary of the CC-OPF problem, and hence is the satisfactory numerical optimum of ICC-OPF.

B. Visualized Analysis of Security Trade-Offs

To visualize the operational trade-offs among the security levels, we perform a case study on the IEEE 14-bus system. The analysis focus on the two-dimensional security level space defined by two critical transmission lines (4,9) and (5,6), while fixing the security levels for all other lines and generators at a constant 0.98. The general model (3d) is specified using an angle $\theta \in [0, \pi/2]$, which defines the direction components $u_{(4,9)} = \cos \theta, u_{(5,6)} = \sin \theta$, and the corresponding offsets $\beta_{0,(4,9)} = \beta_{0,(5,6)} = 0.90$. All other constraints have $u_k = 0$ and a baseline of $\beta_{0,k} = 0.98$, leading to the specific formulation:

$$\beta_k = \begin{cases} \beta \cos \theta + 0.90, & \text{if } k = (4, 9) \\ \beta \sin \theta + 0.90, & \text{if } k = (5, 6) \\ 0.98, & \text{otherwise} \end{cases} \quad (18)$$

This formulation allows us to explore the entire boundary of the feasible security region for lines (4,9) and (5,6) while maintaining a high security level of 0.98 for all other constraints. By repeatedly solving for the maximum security level $\beta_{\max}$ at different values of θ , we trace this boundary as depicted by the blue curve in Fig. 3(a). Any security level pair $(\beta_{(4,9)}, \beta_{(5,6)})$ within this boundary (the lower-left region) is achievable by the system, whereas any pair outside it is not.

The security region boundary reveals a negative correlation between the maximum achievable security levels $(\beta_{(4,9)}^{\max}, \beta_{(5,6)}^{\max})$. This reveals a fundamental trade-off phenomenon that the security level for one critical line is enhanced at the cost of the security level for another, and hence the

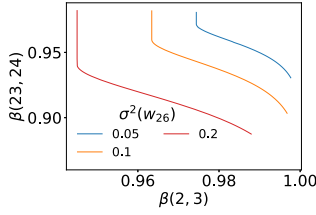


Fig. 4. Security region contracts as uncertainty increases.

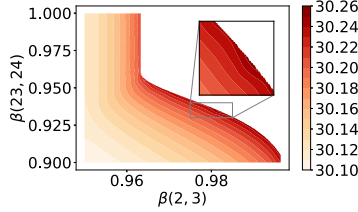


Fig. 5. Operational cost landscape across the security region.

security levels cannot be arbitrarily predefined. This observation underscores the necessity of the ICC-OPF problem, as it provides a systematic way to examine the interdependencies between the security levels of those “competing” constraints.

Furthermore, a sensitivity analysis on the baseline security of the non-prioritized constraints (i.e., those with a baseline of 0.98), as shown in Fig. 3(a). The security region contracts as β_0 is increased, quantitatively illustrating how a more stringent global security requirement reduces the feasible operating space.

The above analysis is repeated for the IEEE 39-bus system, by selecting critical lines (2,3) and (23,24) under the same settings for $\mathbf{u}$ and $\beta_{0,k}$. The resulting security region boundary, shown in Fig. 3(b), reveals a more complex relationship. Unlike the 14-bus case, the boundary is no longer convex; there are regions where decreasing $\beta_{(23,24)}^{\max}$ does not imply an increase in $\beta_{(2,3)}^{\max}$. This non-convex interplay highlights the intricate interactions among chance constraints, further reinforcing the need for a quantitative and systematic approach rather than heuristic setting for security levels.

C. Risk and Economic Analysis for Operations

The security region boundary derived by our method provides a powerful tool for operational planning, visualizing the trade-offs between security and economics under uncertainty.

For instance, if extreme weather increases a wind farm’s output uncertainty (modeled by scaling the variance and covariance of the injection at bus 26) in the IEEE 39-bus system, the security region contracts, as shown in Fig. 4. This quantitatively informs an operator that maintaining a 98% security for line (2,3) now requires reducing the maximum security for line (23,24) from 96% to 90%. This enables direct, non-heuristic re-coordination of security targets.

Furthermore, Fig. 5 reveals the associated economic impact. As expected, operational cost rises with stricter security levels. Notably, the cost contours are parallel to the security boundary, as both are dictated by shared physical limits like line congestion. The varying cost along the boundary also reflects complex economic trade-offs between securing different assets, providing a more informative basis for optimal dispatch decisions.

V. CONCLUSION

This letter formulates the ICC-OPF to evaluate the maximum security levels a power system can achieve under uncertainty. A sensitivity-based NR-like algorithm is designed for efficient computation. Case studies demonstrate the ICC-OPF’s ability to systematically quantify the trade-offs and interdependencies between the security levels of different generators and line flows.

REFERENCES

- [1] D. Bienstock, M. Chertkov, and S. Harnett, “Chance-constrained optimal power flow: Risk-aware network control under uncertainty,” *SIAM Rev.*, vol. 56, no. 3, pp. 461–495, 2014.
- [2] A. Hassan, R. Mieth, M. Chertkov, D. Deka, and Y. Dvorkin, “Optimal load ensemble control in chance-constrained optimal power flow,” *IEEE Trans. Smart Grid*, vol. 10, no. 5, pp. 5186–5195, Sep. 2019.
- [3] Y. Song, T. Liu, and D. J. Hill, “Chance constrained economic dispatch considering the capability of network flexibility against renewable uncertainties,” *IEEE Trans. Power Syst.*, vol. 39, no. 5, pp. 6636–6648, Sep. 2024.
- [4] A. Zhou, M. Yang, M. Wang, and Y. Zhang, “A linear programming approximation of distributionally robust chance-constrained dispatch with Wasserstein distance,” *IEEE Trans. Power Syst.*, vol. 35, no. 5, pp. 3366–3377, Sep. 2020.
- [5] T. Mühlpfordt, T. Faulwasser, L. Roald, and V. Hagenmeyer, “Solving optimal power flow with non-Gaussian uncertainties via polynomial chaos expansion,” in *Proc. IEEE Conf. Dec. Control*, 2017, pp. 4490–4496.
- [6] T. Mühlpfordt, L. Roald, V. Hagenmeyer, T. Faulwasser, and S. Misra, “Chance-constrained AC optimal power flow: A polynomial chaos approach,” *IEEE Trans. Power Syst.*, vol. 34, no. 6, pp. 4806–4816, Nov. 2019.
- [7] G. Chen, H. Zhang, and Y. Song, “Chance-constrained DC optimal power flow with non-Gaussian distributed uncertainties,” in *Proc. IEEE Power Energy Soc. Gen. Meeting*, 2022, pp. 1–5.
- [8] Z. Wang, C. Shen, F. Liu, X. Wu, C.-C. Liu, and F. Gao, “Chance-constrained economic dispatch with non-Gaussian correlated wind power uncertainty,” *IEEE Trans. Power Syst.*, vol. 32, no. 6, pp. 4880–4893, Nov. 2017.
- [9] A. M. Hou and L. A. Roald, “Chance constraint tuning for optimal power flow,” in *Proc. Int. Conf. Prob. Meth. Appl. Power Syst. (PMAPS)*, 2020, pp. 1–6.
- [10] A. M. Hou and L. A. Roald, “Data-driven tuning for chance-constrained optimization: Two steps towards probabilistic performance guarantees,” *IEEE Control Syst. Lett.*, vol. 6, pp. 1400–1405, 2021.
- [11] D. Bienstock and A. Shukla, “Variance-aware optimal power flow,” in *Proc. Power Syst. Comput. Conf. (PSCC)*, 2018, pp. 1–8.
- [12] D. Bienstock and A. Shukla, “Variance-aware optimal power flow: Addressing the tradeoff between cost, security, and variability,” *IEEE Trans. Control Netw. Syst.*, vol. 6, no. 3, pp. 1185–1196, Sep. 2019.
- [13] F. Oldewurtel, L. Roald, G. Andersson, and C. Tomlin, “Adaptively constrained stochastic model predictive control applied to security constrained optimal power flow,” in *Proc. Amer. Control Conf.*, 2015, pp. 931–936.
- [14] A. Ghosh, C. Cortes-Aguirre, Y.-A. Chen, A. Khurram, and J. Kleissl, “Adaptive chance constrained MPC under load and PV forecast uncertainties,” in *Proc. IEEE PES Grid Edge Technol. Conf. Expo. (Grid Edge)*, 2023, pp. 1–5.
- [15] A. Ghosh, C. Cortes-Aguirre, Y.-A. Chen, A. Khurram, and J. Kleissl, “Adaptive relaxation-based nonconservative chance constrained stochastic MPC,” *IEEE Trans. Control Syst. Technol.*, vol. 33, no. 5, pp. 1543–1559, Sep. 2025.
- [16] T. C. Chan, R. Mahmood, and I. Y. Zhu, “Inverse optimization: Theory and applications,” *Oper. Res.*, vol. 73, no. 2, pp. 1046–1074, 2025.
- [17] Q. Li, Z. Liang, A. Bernstein, and Y. Dvorkin, “Revealing decision conservativeness through inverse distributionally robust optimization,” *IEEE Control Syst. Lett.*, vol. 8, pp. 1018–1023, 2024.
- [18] A. J. Conejo, E. Castillo, R. Mínguez, and R. García-Bertrand, *Decomposition Techniques in Mathematical Programming: Engineering and Science Applications*. Heidelberg, Germany: Springer, 2006.
- [19] R. D. Zimmerman, C. E. Murillo-Sánchez, and R. J. Thomas, “MATPOWER: Steady-state operations, planning, and analysis tools for power systems research and education,” *IEEE Trans. Power Syst.*, vol. 26, no. 1, pp. 12–19, Feb. 2011.